Analytical Chemistry Cumulative Exam

February 16, 2024

"Back to Basics" The role of Equilibrium in Analytical Chemistry.

1) Cyanide (CN^-) is an important analyte to quantify for many different reasons.[1] While CN^- ion-selective electrodes (ISE) are still plagued with interference problems, we will assume we have found the ideal ISE that detects CN^- without interference.

HCN has a pK_a of 9.2.

a) If you are making standards of 10 mM NaCN in a borate buffer, pH 8.0, what is the concentration of CN^- that your ISE will report (assume you have it calibrated to know how the voltage relates to concentration)?

b) If you made the 10 mM NaCN standards in polished water (e.g., Nanopure or HPLC grade) would the sensor give a reading that is equivalent to 10 mM NaCN? Why or why not?

2) You have water samples with unknown concentrations between 10 and 25 mM for each ion of OH^-, CO_3^{2-}, and HCO_3^-.

For water, $K_w = 10^{-14}$ For H_2CO_3, $pK_{a1} = 6.4$ and $pK_{a2} = 10.3$.

Describe how you would quantify these three ions within this unknown sample. Very clearly denote what information you must obtain to determine the concentration of each of these in the unknown sample. If any spreadsheet process is needed, explain what must be coded or performed with the spreadsheet calculations.

Assume you can easily access >100 mL of sample and have access to any wet chemistry techniques that you need, e.g., glassware, strong acids or bases, balances, and pH meters.

3) You have two separate analytes in an aqueous sample that you need to completely remove from each other. You choose to perform a solvent extraction where the partition coefficient for both neutral forms of the analytes is $\mathbf{K_p=1000}$ between hexane and water.

One analyte is an organic carboxylic acid (R-COOH) with a **pK_a of 5.**

The other analyte is an organic amine ($R\text{-}NH_3^+$) with a **pK_a of 9.**

Explain how you would perform this extraction such that both analytes are at least >90% removed from each other. Justify your choices.

You have full access to a well-equipped laboratory with glassware, acids/bases, and so forth.

[1] I don't expect you to know a lot about cyanide, but if interested here is a comprehensive review about its analytical chemistry. https://doi.org/10.1016/j.aca.2010.05.042 *Recent developments in cyanide detection: A review.* Jian Ma and Purnendu K. Dasgupta